# Well Interference and Fracture Geometry Investigation Using Production and Low-Frequency Distributed Acoustic Sensing Data in an Unconventional Reservoir

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## Summary

Well interference directly impacts well spacing optimization and well economics, thus it has received significant attention. Observed in many unconventional reservoirs, well interference is mainly attributed to connection of hydraulic fractures. Few studies have compared the hydraulic fracture connection during stimulation with that during the production period. This study connects these two aspects by leveraging the low-frequency distributed acoustic sensing (LF-DAS) signals acquired during the stimulation and the daily production data of 23 horizontal wells in the Denver-Julesburg (DJ) Basin. Strong cross-formation communication between Niobrara and Codell has been identified from both LF-DAS and production data. Interwell fracture connection could divert up to 80% of a shut-in well's production to the offset well in the adjacent formation, revealing that well interference can be evaluated through random shut-ins of wells during production. Further, fracture connection in LF-DAS data is observed between Codell wells but rarely between Niobrara wells. Given the average Codell well spacing of 1,270 ft and Niobrara well spacing of 650 ft, we observe that the half-length of the stimulated hydraulic fractures in Codell wells is significantly longer than that of Niobrara wells. This study also confirms that stronger well interference during production correlates with strong fracture connection identified from LF-DAS.

## Introduction

Multistage hydraulic fracturing stimulation is a standard technique to recover hydrocarbon from unconventional reservoirs. Appropriate well spacing is an important contributing factor to extract more hydrocarbon in place with minimal cost. From 2009 until recently, when targeting the Niobrara and Codell formations in Denver-Julesburg (DJ) Basin, well densities have increased from 1 horizontal well per section to 11–15 horizontal wells per section (Milne and Cumella 2014; Anderson et al. 2015; Ning et al. 2017). As more wells are drilled, wells tend to communicate with their neighbors and well interference has been observed in many unconventional plays such as Haynesville shale, Marcellus shale, and Eagle Ford (Khodabakhshnejad 2019; Gupta et al. 2020; Raterman et al. 2020). In the Eagle Ford, a well spacing of 400–800 ft is likely to facilitate communication between offsets (Raterman et al. 2019). It is essential to understand well interference because it directly affects well spacing optimization and thus impacts the economic development of unconventional reservoirs (Portis et al. 2013; Yaich et al. 2014; Lindner and Bello 2015; Rimedio et al. 2015).

Well interference typically includes two categories: hydraulic fracture connection during completion and fluid communication during production. When a nearby well is being hydraulically fractured, pressure increase at an offset well can be detected using pressure monitoring gauges (Dawson and Kampfer 2016; Raterman et al. 2019); location of microseismic events can be assessed (Fisher et al. 2004); fracture hits can be observed using the fiber-optic-based DAS measurement (Jin and Roy 2017); and stepwise water rate increase can be measured from a neighboring well that is in production. During initial production, radioactive tracers can be used to identify the interwell flow communication (Ning et al. 2017). During long-term production, production rate change caused by an offset well's operation can be clearly observed from the production decline curves (Ajani and Kelkar 2012). Wells are interconnected through the hydraulic fractures. When one well is shut-in or experiencing other operations, the pressure gradient allows hydrocarbon to flow toward offset wells. Hence, an increase in oil/gas production from offset wells is observed.

Well interference is due to the connection of hydraulic fractures, which is determined by geologic properties, geomechanics, and completion strategies. Based on Rassenfoss (2020), all four wells in the Eagle Ford pilot test were interconnected by hydraulic fractures during the first 60 days of production, but the connections between wells had deteriorated after 400 days of production. Raterman et al. (2019) concluded that assuming uniform fracture geometries cannot adequately represent well interference. The distributed temperature sensing warm-back data, pressure data, and integrated modeling results support nonuniform two-region fracture geometries—a more intensely drained near-well region of secondary fractures surrounded by an outer region of irregularly distributed principal fractures. Secondary fractures of the more homogeneous inner region are located between clusters and have a relatively shorter length. They could be highly conductive and are crucial for drainage in the near-wellbore region. The conductivity decreases away from the producing lateral. The principal fractures of the outer region can grow more than 1,000 ft during stimulation (Raterman et al. 2020) in some formations, but they are largely ineffective for drainage beyond several hundred feet of the producing wellbore. Based on that, Acuña (2018, 2020, 2021) used the fracture swarm model with many closely spaced fractures to understand unconventional well behavior and explain those apparently unrelated observations.

In the DAS measurement, an optical fiber cable is deployed along the monitor wellbore. When the laser energy is injected into the fiber, each piece of the fiber naturally scatters a small portion of the laser energy back to the surface-sensing unit. The surface-sensing unit then

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measures the strain change along the fiber using the interferometry technique. The strain change measurement has meter level spatial resolution, which is critical in detecting and monitoring the near-wellbore activities.

The raw DAS data are usually sampled at a very high temporal rate (5000–10,000 Hz), which can be processed and downsampled to a low frequency (<0.5 Hz). The LF-DAS measurement can capture the slow strain changes influenced by temperature and mechanical strain along the fiber. LF-DAS has been an emerging and effective technology to identify fracture height, length, and density during completion. During hydraulic fracturing operation, when the fractures grow and hit the monitor fiber in the offset horizontal well, extensional strain can be observed at fracture hit locations, which indicate fracture opening. The extensional strain is usually represented by red color in the data virtualization. Correspondingly, the compressional stress shadow on both sides of the fractures is represented by blue color. After the pumping operation stops, the fiber at fracture hit locations is compressed due to the closing of the newly created hydraulic fractures (Jin and Roy 2017). By analyzing a monitoring well's LF-DAS data during the hydraulic fracturing operation of its offset producers, Jin and Roy (2017) observed three fractures in a stage that propagated at least 700 ft horizontally; Raterman et al. (2018) identified that some fractures extend for 1,500 ft in one Eagle Ford acreage. These fracture hit observations correspond to principal fractures in the nonuniform two-region fracture geometries.

Although previous studies have investigated well interference, either from the production side or during the stimulation period, few studies have connected and compared the two. This study leverages the two sides to constrain hydraulic fracture geometries in the four-square-mile study area of DJ Basin, as well as to assess the interference among the wells located in the Niobrara and Codell formations. The 23 horizontal wells and geological properties will be discussed first, followed by the analysis of production interference and the investigation of LF-DAS results recorded from two representative Niobrara and Codell wells. The production interference observation was then compared with LF-DAS fracture hit results to better understand fracture connection of neighboring wells. The objective of this study is to estimate the hydraulic fracture geometry, better understand the conductivity change of long hydraulic fractures created during stimulation, as well as provide insights for future stimulation design in this field.

## Study Area

As illustrated by the small dark blue square in **Fig. 1,** the Chalk Bluff study area is located in the Hereford Field of the northern DJ Basin, close to the Wyoming-Colorado state line. The currently producing formations are Niobrara and Codell **(Fig. 1)**. The Niobrara formation has a thickness of about 300 ft. It consists of multiple members: A Chalk, A Marl, B1 Chalk, B1 Marl, B Chalk, B Marl, C Chalk, C Marl, and Fort Hays. Chalk units are considered as the reservoir rock while marls containing higher total organic carbon contents are regarded as the source beds (Sonnenberg 2016). Though all members of Niobrara contribute to production, Niobrara B Chalk is the drilling target zone in this study, which has a thickness of 25–30 ft. Some stages of Niobrara well laterals make a slow transition into the adjacent B Marl and B1 Marl. For Niobrara, the porosity is about 5–8% and permeability is approximately $10^{-6}$ to $10^{-5}$ md. The underlying Codell sandstone is a second drilling target zone. It has a thickness of 18–20 ft. When compared with Niobrara, Codell has higher porosity (9–12%) and permeability ($10^{-3}$ to $10^{-2}$ md).

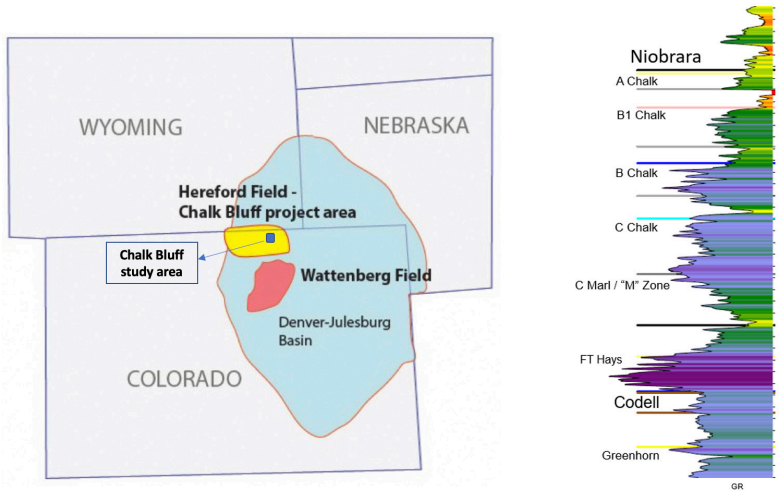


**Fig. 1—The four-square-mile Chalk Bluff study area is located in the north DJ Basin (left). Niobrara and Codell are the two producing zones (right).**

In the four-square-mile study area, 23 horizontal wells were drilled in 2019 following a north-south direction. These wells were separated into four pads, as illustrated by different colors in **Fig. 2a.** Twenty-two wells have the lateral length of approximately 9,900 ft while the remaining one well (P2Df) has a shorter lateral length of about 7,500 ft. The "P2" in the well name "P2Df" means this well is located on Pad 2. Codell wells are marked by dashed lines, whereas Niobrara wells are represented by solid straight lines. **Fig. 2b** shows the gun barrel diagram of the 23 wells. The *y*-axis is true vertical depth (TVD). The *x*-axis shows the horizontal distance to the westernmost well, P4F. Each well is symbolized by its toe location. Niobrara wells are shaped by circles. The underlying Codell wells are shaped by square boxes. Seven wells are Codell wells, and the remaining are Niobrara wells. Niobrara wells have a spacing ranging from 440 to 1,000 ft, among which the majority have a spacing of 650 ft. Codell wells have a larger spacing of 1,270–2,400 ft. The LF-DAS signals acquired from two fiber wells were investigated in this study—the Codell well P3Bf and the Niobrara well P3Df. The data used in this study include LF-DAS (below 0.5 Hz) signals of the two fiber wells, detailed completion data of seven wells on Pad 3, and daily production data of all 23 wells.

## Production Interference Analysis

The daily production data of all 23 horizontal wells are available in this study. The operation of occasional shut-in and water fracture hits led to well interference. To quantify the impact of well interference, the decline curve was estimated for each well using the production

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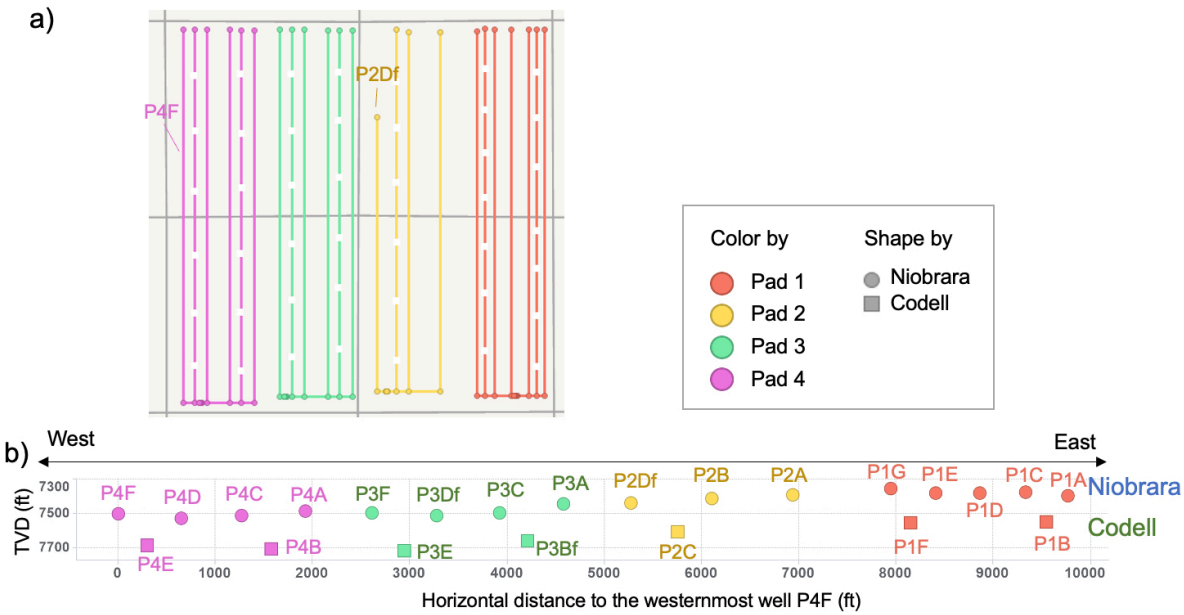


**Fig. 2—(a) The four-square-mile study area includes 4 pads and 23 horizontal wells; (b) in the gun barrel diagram, each dot represents one well. The wells are colored by pad number and shaped by circles for Niobrara and squares for Codell formations. Notice wider Codell spacing vs. Niobrara. The vertical scale is exaggerated.**

data of the undisturbed period. The curve is matched using the Arps equation as shown in Eq. 1. Here, $q_t$ is production rate at time $t$ (B/D), $q_i$ is the initial production rate (B/D), $b$ is the dimensionless decline exponent, $D_i$ is the initial decline rate (days$^{-1}$), and $t$ is time (days). Least-squares minimization is used to estimate the optimal parameters in Eq. 1 to match the real production data.

$$q_t = \frac{q_i}{\left(1 + bD_i t\right)^{\frac{1}{b}}}. \tag{1}$$

Among the four pads in the study area, the production curves of wells on Pads 1, 2, and 3 clearly show an increasing trend when the shut-in or similar operations occurred at neighboring wells. This section will investigate the production data of the wells on the three pads.

**Well Interference Caused by the Shut-In Operation. Fig. 3b** plots the wells on Pad 3 and the neighboring two wells on Pad 4. The relative location is plotted in a gun barrel diagram. Regardless of targeting zones, the wells are named P3A, P3B, P3C, P3D, P3E, and P3F from east to west. Four wells are in Niobrara while the remaining two are Codell wells (P3Bf and P3E). The four Niobrara wells have a spacing of approximately 450–700 ft, and the two Codell wells are about 1,270 ft apart. Production of all seven wells started in August 2019.

**Fig. 3a** shows the production curves of P3Df, P3E, P3F, and P4B. On the production plots, the green dots represent real production from August 2019 to January 2021. According to the production behavior, the whole production history was divided into four stages: Stage 1 (1 August 2019–1 March 2020) when the choke size changes over time; Stage 2 (1 March 2020–10 April 2020) when choke size remained constant and production declines following the Arps equation; Stage 3 (10 April 2020–1 August 2020) when abnormal production rates occur due to operation or well interference; and Stage 4 (1 August 2020–10 January 2021) when production gets back to the original decline curve. In Stage 4, well interference may still exist, but we could not evaluate it due to the disappeared production disturbance. The production data in Stage 2 and Stage 4 are collected to fit Eq. 1 with a goal of estimating the Arps parameters to achieve a production match. The blue dots and the orange dots represent the production values that were calculated by the estimated Arps equations.

During the shut-in period of Well P3E, the nearby Codell well P4B's production continued following the decline curve, revealing P4B is not disturbed by P3E shut-in. The gas lift system in another neighboring Codell well, P3Bf, was lowered around April 2020. Thus, it is difficult to distinguish whether P3Bf is influenced by the shut-in operation of P3E. All these observations are summarized by the black dash arrows in **Fig. 3b**.

During Stage 3, when P3E was shut in on 10 April 2020, P3F showed continuous production increase after that day, as indicated by the blue arrow in **Fig. 4a.** It shows that P3F's production was affected by P3E shut-in. P3Df's oil rate showed a weak disturbance for 2 days, which was then interrupted by the operation of lowering gas lift system and dropping plungers. The production disruption of P3Df was followed with a short buildup, which could be a function of coming to equilibrium with the new operational setup. Because P3Df had its own operation during the same time period, it is difficult to attribute P3Df's production change to the shut-in operation of P3E.

On 2 May 2020, P3E was reopened and showed temporary high oil production. Accordingly, the oil rate of P3F showed a sudden drop from 95 B/D on 2 May 2020 to 85 B/D on 4 May 2020, and then kept decreasing. This further validates the production communication between P3E and P3F. P3Df showed erratic oil rate, which might be caused by noise or the continuing effect of P3Df's own operation. It cannot demonstrate production communication from P3E to P3Df, as illustrated by the question mark in **Fig. 4b**.

As mentioned above, P3E affected the production of P3F. Due to the operation synchronization of P3Df and P3E, no available data can establish the impact of P3Df on P3F. This is represented by the question mark in **Fig. 4b**. Hence, it is assumed that P3F's production increase is fully related to P3E. **Table 1** quantitively summarizes the production loss caused by the shut-in operation of P3E and the



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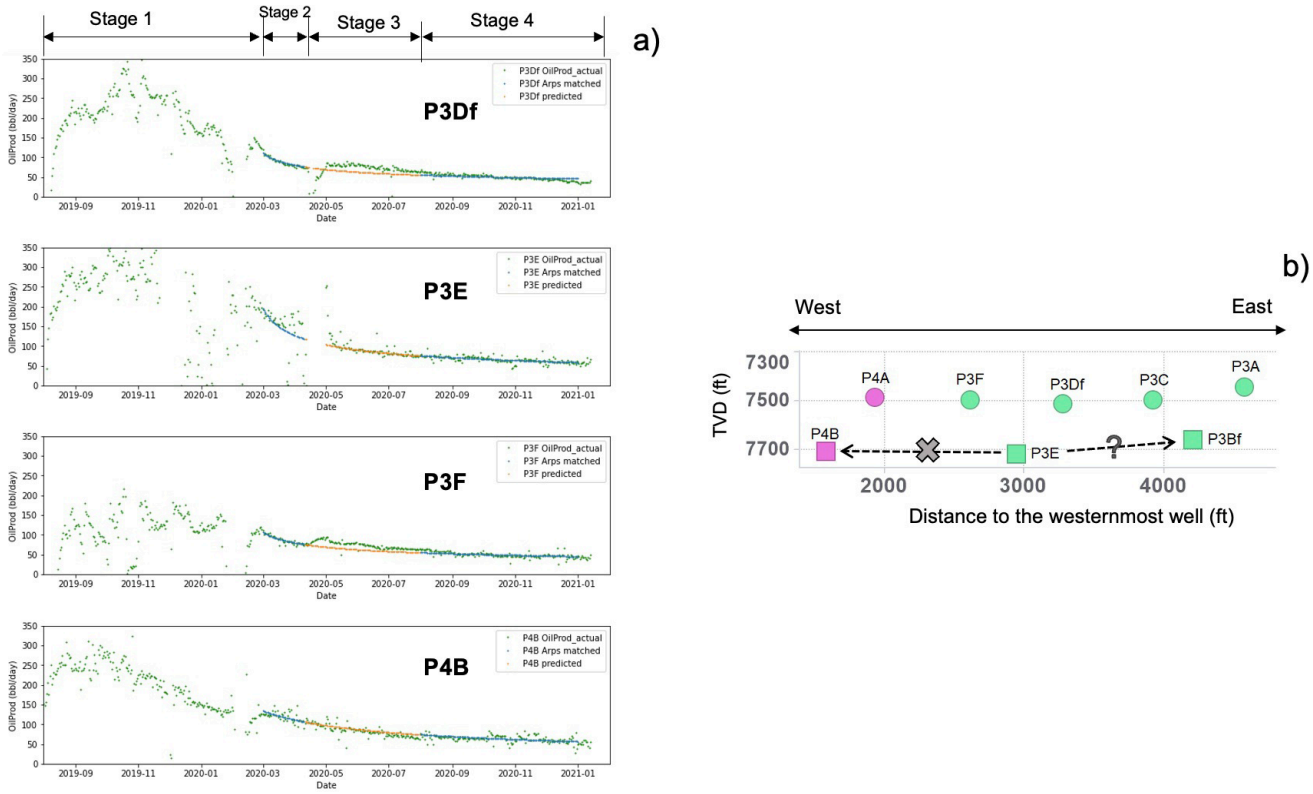


**Fig. 3—(a) Shut-in operation of P3E improves P3F's production but does not affect P4B, which is located in the same formation; (b) there are four Niobrara wells and two Codell wells on Pad 3. No production communication within Codell is observed.**

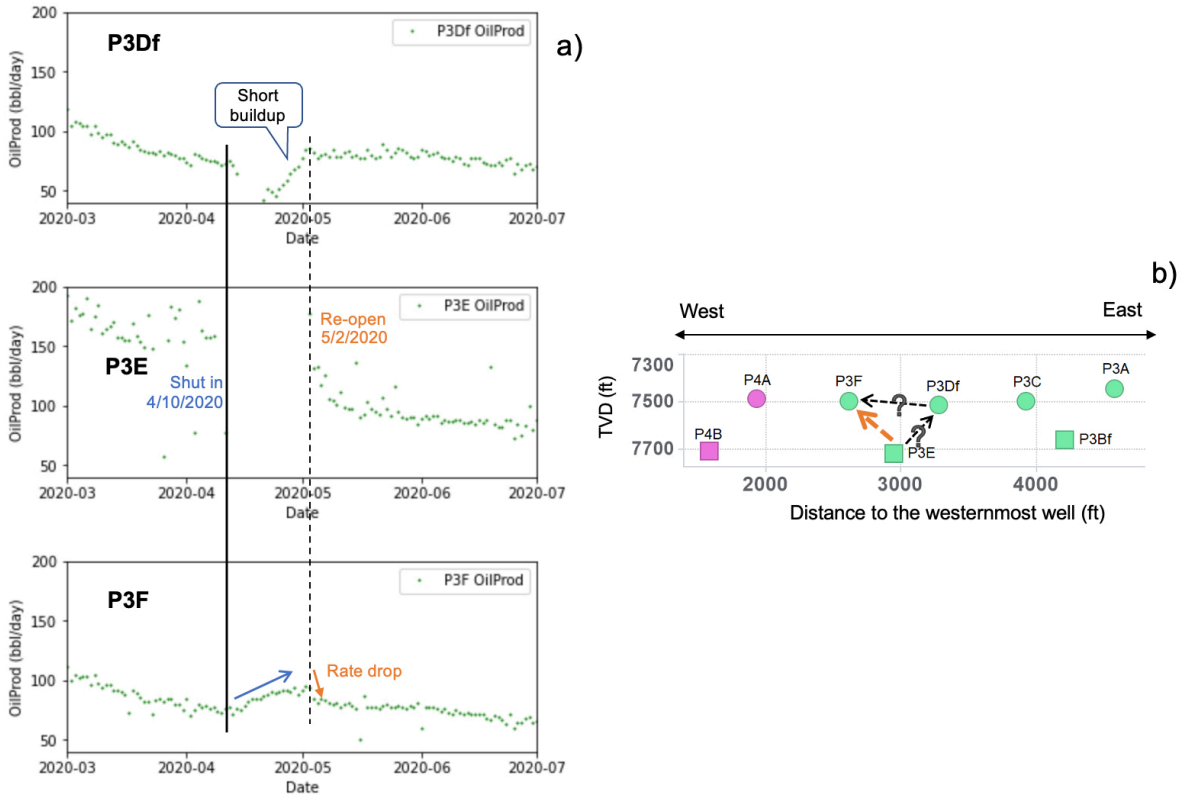


**Fig. 4—(a) The production increase and drop of P3Df and P3F are related to P3E's shut-in and reopen operations. (b) Production communication existed from P3E to P3F.**

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| Well (Zone) | Changed Oil Production (bbl) in 112 Days (10 April 2020–1 August 2020) | Equivalent to B/D | Shifted Ratio |
|---|---|---|---|
| P3E (Codell) | −1,536.92 | −13.72 | – |
| P3F (Niobrara) | 1,228.04 | 10.96 | 80% |

Table 1—An 80% of P3E's production loss is diverted to its neighboring well P3F.

production increase of P3F caused by well interference. The production change in Stage 3 is estimated by the actual production rate (green dots in **Fig. 5**) subtracting the values predicted by the Arps equation (orange dots). The shifted ratio is defined as the increased production of P3E divided by the production loss of P3F. The quantified production impact is calculated and summarized in **Table 1**.

**Well Interference Initiated by Water Fracture Hits.** Wells on Pad 1 and Pad 2 did not have shut-in operations that are suitable for well interference analysis. However, some wells' production was affected by the water fracture hit of the offset wells outside the study area. Though the water fracture hit impact is less ideal than the shut-in operation from the well interference analysis perspective, it still provides valuable insights to investigate production interference.

As illustrated in **Fig. 6,** five Niobrara wells (Nio1, Nio2, Nio3, Nio4, and Nio5) were completed by March 2020. During the stimulation of the wells Nio1–Nio5, due to the fracture azimuth of N55°E in Niobrara, the stimulated water flows to the toe stages of the study wells (P1A, P1C, and P2B), which results in the bottomhole pressure increase. The toe stages connected to the hydraulic fractures full of stimulation water and with high pressure are prioritized to drain, resulting the water production increase as observed. Accordingly, the oil production of remaining stages is suspended. Hence, the sudden decrease of oil rate and gas rate is observed from the wells P1A, P1C, and P2B during the first quarter of 2020. This generates a similar pressure perturbation within the fracture system as shut-in of the fracture hit wells, as pressure is less depleted by the wells due to temporal borehole pressure increase.

***Production Disturbance on Pad 1.*** As shown in **Fig. 5b**, Pad 1 has five Niobrara wells and two underlying Codell wells (P1B and P1F). Niobrara wells have a spacing of approximately 450 ft while the two Codell wells are 1,400 ft apart. Production of all seven wells started in June 2019.

Well interference is observed from the production data of wells P1A, P1B, and P1C. **Fig. 5a** shows the production curves of P1A, P1B, and P1C. The oil rate decrease of wells P1A and P1C in Stage 3 is attributed to the water fracture hits of the northeast Niobrara wells stimulated in Q1 2020 outside the study area, while P1B was not affected by the water fracture hits (please refer to Fig. S-1 in the Supplementary Material for more details). Stage 3 of the three production curves in **Fig. 5a** shows that when the actual production (green dots) of P1A and P1C is lower than the values predicted by the decline curves (orange dots), the offset well P1B shows increased production in a couple of days. This observation demonstrates strong production communication between Niobrara and Codell wells.

The next step is to study if P1B's production increase is related to P1F, which is the adjacent well in the same formation, as illustrated in **Fig. 5b**. When P1B's production started increasing after 25 January 2020, no abnormal production rate can be observed at P1F, indicating P1B's production increase at Stage 3 is not related to P1F.

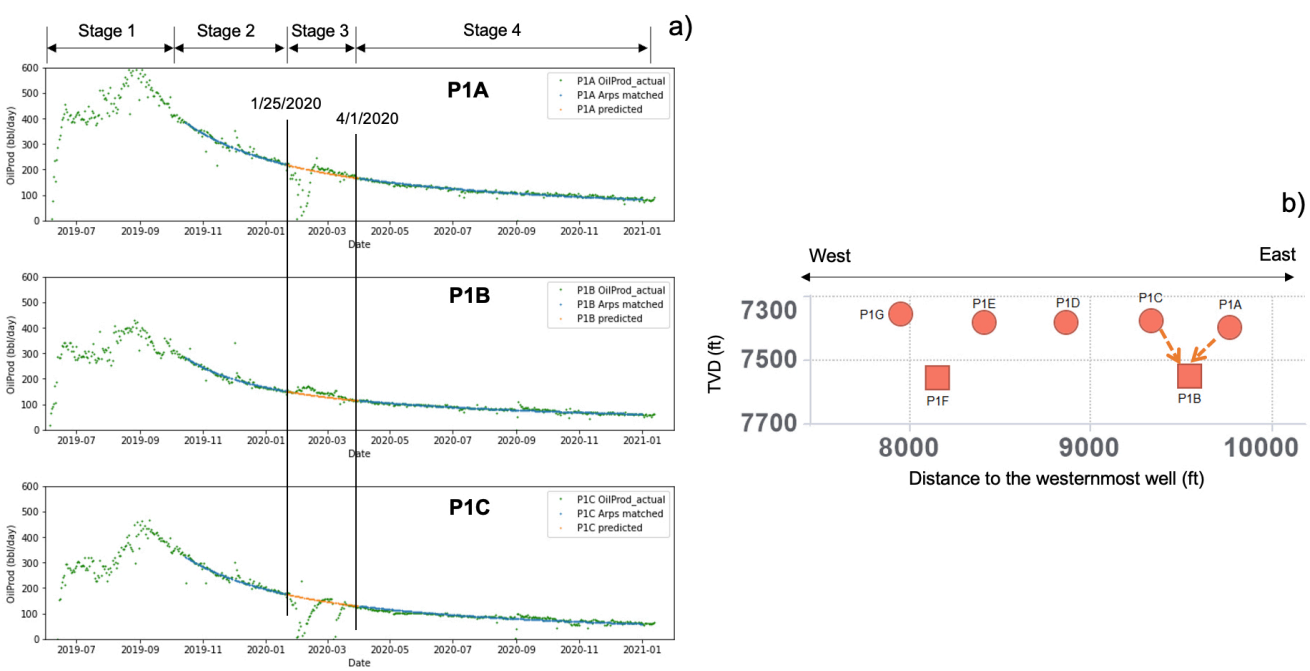


**Fig. 5—(a) The Codell well P1B's production increase is impacted by the production decrease of the neighboring Niobrara wells P1A and P1C. (b) There are five Niobrara wells and two Codell wells on Pad 1.**



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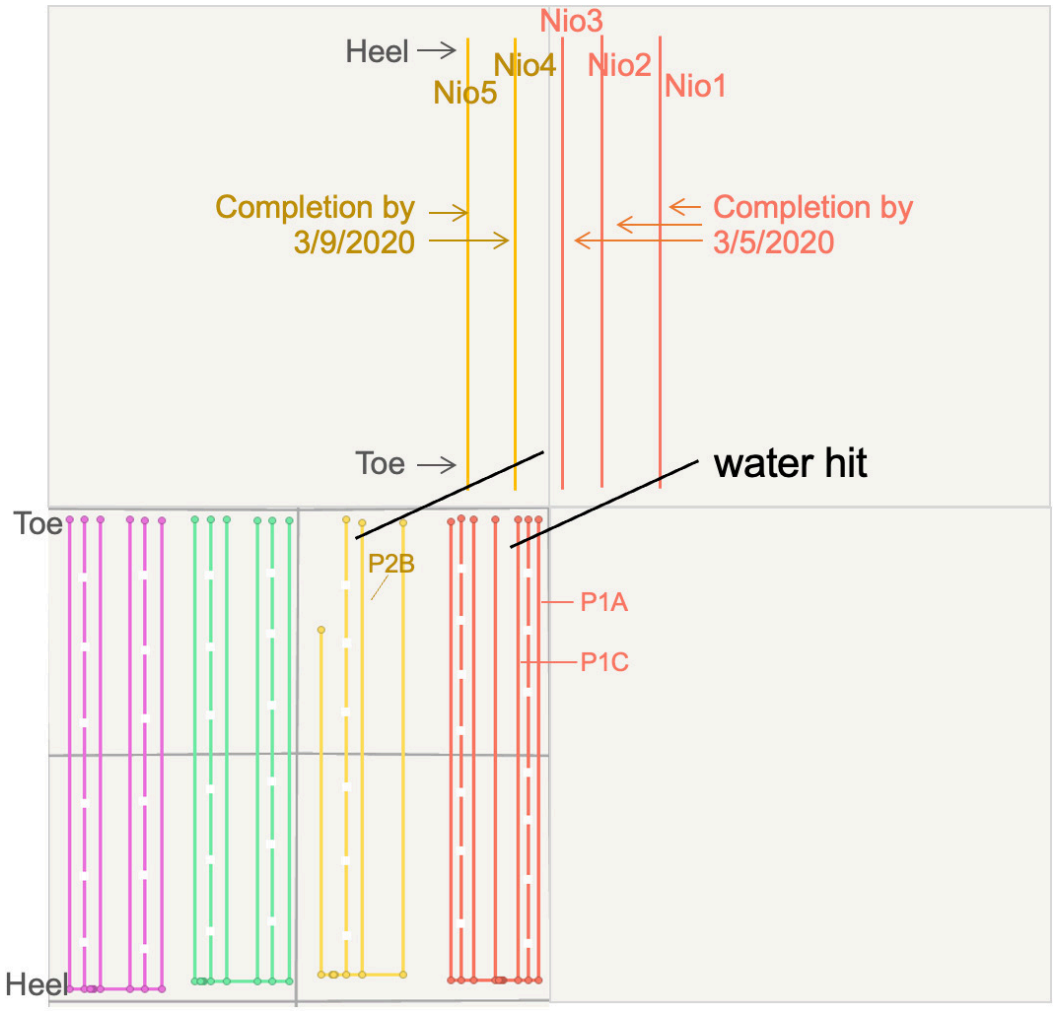


**Fig. 6—Water hit of wells outside the study area (Nio1–Nio5) caused the increased water cuts and decreased oil and gas production in wells on Pad 1 and Pad 2. Black lines illustrate that the water fracture hits from Nio1 to Nio5 lead to the production change of wells within the study area.**

**Table 2** quantitively summarizes the production losses of P1A and P1C and the production increase of P1B caused by well interference. The shifted ratio is defined as the increased production of P1B divided by the total production loss of P1A and P1C. Here it is assumed that the production disturbance of both P1A and P1C results in the rate increase of P1B. The reason behind this is that P1A and P1C have very similar well spacing to P1B and have the synchronization of production decrease.

| Well (Zone) | Changed Oil Production (bbl) in 66 Days (25 January 2020–01 April 2020) | Equivalent to B/D | Shifted Ratio |
|---|---|---|---|
| P1A (Niobrara) | −1,498.29 | −22.7 | – |
| P1B (Codell) | 725.06 | 10.99 | 18% |
| P1C (Niobrara) | −2,558.66 | −38.77 | – |

Table 2—Comparison of production loss and increase of P1A, P1B, and P1C.

***Production Disturbance on Pad 2.*** **Fig. 7b** plots the four wells on Pad 2 and their neighboring wells on Pads 1 and 3. The wells on Pad 2 are named P2A, P2B, P2C, and P2D from east to west. Three wells are located in the Niobrara formation, and one well is in Codell (P2C). The three Niobrara wells are approximately 800–1,000 ft apart. The Codell well P2C is 2,400 and 1,500 ft away from the offset Codell wells P1F and P3Bf, respectively.

Well interference is observed from the production data of wells P2B, P2C, and P2Df. **Fig. 7a** divides the production curves of P2B, P2C, and P2Df again into four stages, where Stage 3 is the period when well interference occurs. The oil rate decrease of the Niobrara well P2B is caused by the water fracture hit of the Niobrara wells stimulated between February 2020 and March 2020 outside the study area, while P2C was not affected by the water fracture hit (please refer to Fig. S-2 in the Supplementary Material for more details). When



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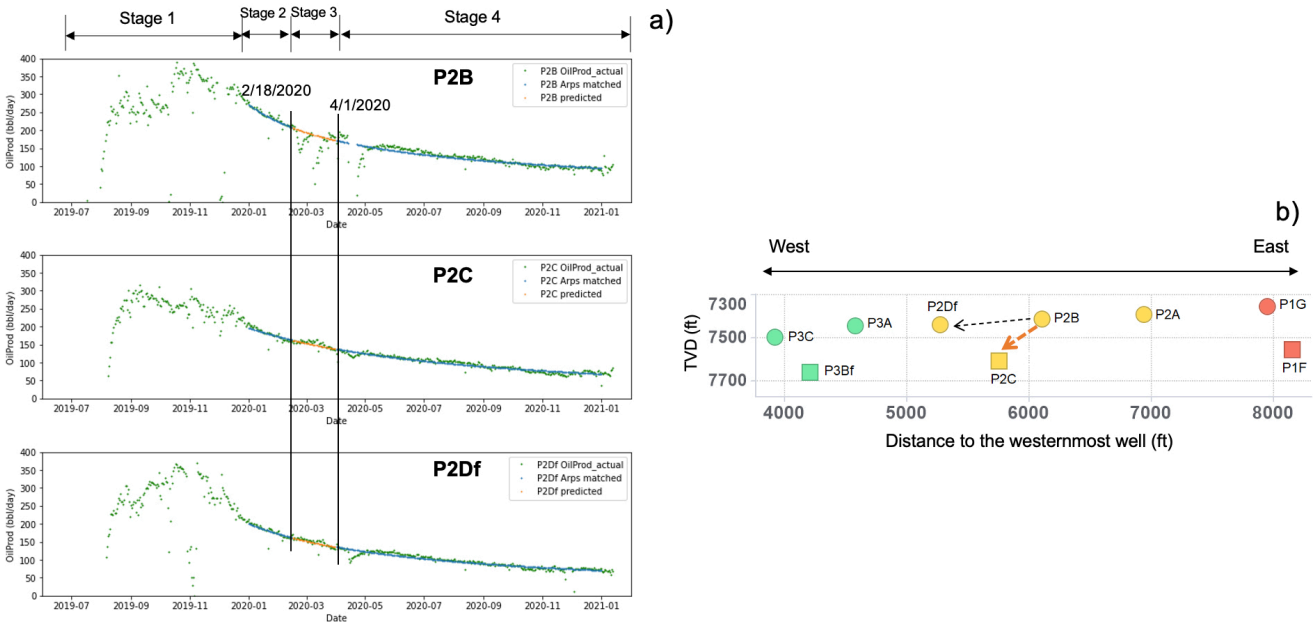


**Fig. 7—(a) Production loss of Niobrara well P2B results in the production increase of P2C. The incremental amount of P2Df's production is negligible. This indicates strong cross-formation communication and negligible communication within Niobrara; (b) there are three Niobrara wells and one Codell well on Pad 2.**

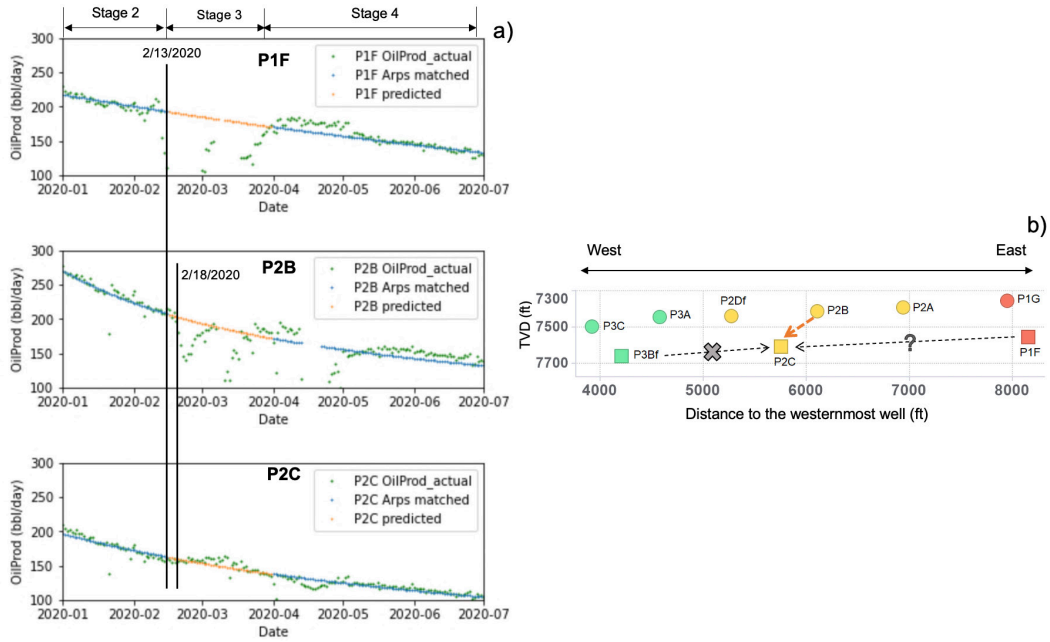


**Fig. 8—P2C's production increase is more affected by P2B than P1F. Cross-formation communication is stronger than the inner formation communication of Codell.**

P2B's production decreases on 18 February 2020, its neighboring Codell well P2C shows production increase within a few days, indicating the interformation communication between P2B and P2C. Another adjacent Niobrara well P2Df also shows slight or negligible incremental oil production, and the increased amount is much less than that of P2C. This reveals that Niobrara-Codell communication is much stronger than Niobrara-Niobrara communication for adjacent wells.

The next step is to distinguish whether P2C's production increase in Stage 3 is related to its Codell neighbors P3Bf and P1F, as presented in **Fig. 7b**. During Stage 3, P3Bf's production declines follow its original trend, meaning P2C's production increase is not related to P3Bf. However, P1F is shut-in during Stage 3, implying P2C's production might be impacted by P1F's operation. To differentiate whether P2C's production increase is from the Codell well P1F or the Niobrara well P2B, the production curves were magnified, as shown in **Fig. 8.** Well P1F's production started decreasing on 13 February 2020, whereas P2B's production decreased 5 days later, 18 February 2020; then, P2C's production increased subsequently. This implies P2C is more likely affected by P2B instead of P1F. **Table 3** quantifies the production interference of wells on Pad 2. When calculating the shifted ratio, it is assumed that P2C's production increase is fully caused by the production disturbance of P2B.



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| Well (Zone) | Changed Oil Production (bbl) in 42 Days (18 February 2020–01 April 2020) | Equivalent to B/D | Shifted Ratio |
|---|---|---|---|
| P2B (Niobrara) | −1,191.70 | −28.37 | – |
| P2C (Codell) | 131.27 | 3.13 | 11% |
| P3Df (Niobrara) | 28.45 | 0.68 | 2% |

Table 3—During the production disturbance of the Niobrara well P2B, Codell well P2C recovered more production than P3Df.

The above quantified production interference results are summarized in **Fig. 9.** Strong communication between Niobrara and Codell wells has been observed. It is estimated that 11–80% of the impacted wells' production was shifted to the offset wells. The values of shifted ratio vary with different operations. Compared to the well disturbance caused by water fracture hits, the shut-in operation leads to the highest shifted ratio (80%). However, Niobrara wells show negligible intraformation communication while Codell wells such as P3E and P4B do not show intraformation communication. As a result, for Codell wells, the sum of the drained hydraulic fracture half-length of P3E and P4B is less than the current spacing of 1,270 ft. When taking an average, the half-length of the drained hydraulic fractures in Codell wells is less than 635 ft.

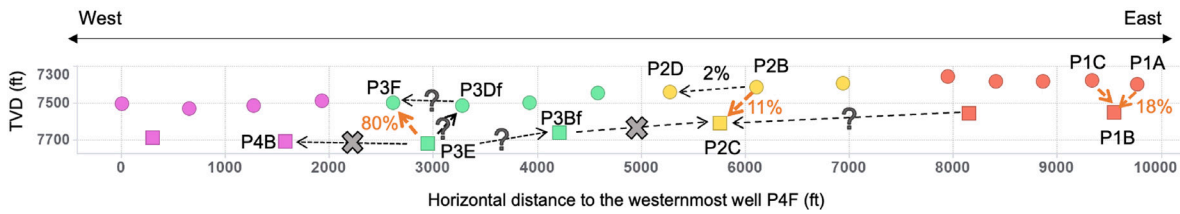


**Fig. 9—Summarization of production interference of all pads. Interformation communication is much stronger than intraformation communication.**

## LF-DAS Data Analysis

This section analyzes the LF-DAS data recorded during the zipper fracturing process of nonfiber wells on Pad 3. As shown in **Fig. 3b**, Pad 3 has four Niobrara wells (P3A, P3C, P3Df, P3F) and two Codell wells (P3Bf, P3E), among which P3Bf and P3Df are the two fiber wells. The LF-DAS measurement of the two wells will be compared and discussed.

**Signals from Niobrara Fiber Monitoring Well P3Df.** The Niobrara fiber well P3Df recorded the LF-DAS signals during the stimulation process of P3E and P3F. A representative example is shown in **Fig. 10. Fig. 10a** indicates that completion fluid was injected into P3E (yellow curve, Codell well) and P3F (black curve, Niobrara well) in a zipper fracturing pattern. Each well (P3E and P3F) has 83 hydraulic fracturing stages with 3 clusters per stage and 40-ft cluster spacing. P3F was completed with 50% more sand and 60% more completion fluid than P3E, thus a longer pumping time was observed, as shown by the black curve in **Fig. 10a**.

**Fig. 10b** presents the LF-DAS signal when P3Df monitors the zipper fracturing of P3F and P3E. During pumping period, red color implies extending strain along the fiber induced by the opening of new fractures. The blue color on both sides of the fractures indicates compressional stress shadow, as marked in **Fig. 10b**. Three groups of fracture hit signals from Well P3E were observed. The fracture hit time is correlated with the injection period of P3E. This indicates that the P3E's fractures reach the Niobrara fiber well P3Df within a short delay after the onset of pumping. In addition to the examples shown in **Fig. 10b**, more fracture hit signals from most P3E injection stages were identified in P3Df, which further demonstrates the cross-formation connections from Codell to Niobrara. Given that P3Df and P3E have a TVD difference of 190 ft, the hydraulic fracture height should be similar or longer than this value. After pumping stops, the newly opened fractures start to close while the compressing area (also called stress shadow, in blue) relaxes immediately. This phenomenon is referred to as polarity flip. It indicates the hydraulic fractures are connected back to the injector P3E. However, no LF-DAS response signal corresponding to the injection of P3F was observed, though P3F is located in the same Niobrara formation with the monitor well P3Df and was injected with 50% more sand and 60% more fluid than P3E. This indicates that the hydraulic fractures from P3F did not reach its adjacent well P3Df in the same formation. Considering the well spacing between P3F and P3Df is about 650 ft, the half-length of P3F's western fractures should be less than this value. **Fig. 10c** summarizes the LF-DAS results that no connection within Niobrara is identified but there is strong communication between Niobrara and Codell.

**Comparison of LF-DAS Signals from Different Fiber Monitoring Wells. Fig. 11** compares the LF-DAS signals recorded from different fiber wells. During the pumping schedule of P3E, **Fig. 11a** shows the LF-DAS signals recorded along the Niobrara fiber well P3Df, whereas **Fig. 11b** presents the LF-DAS signals acquired along the Codell fiber well P3Bf. **Fig. 11c** summarizes the observation of fracture hit from P3E.

As shown in **Fig. 11a**, the fracture hits from the Codell well P3E to the Niobrara fiber well P3Df indicate hydraulic fracture connection exists between Niobrara and Codell. In **Fig. 11b**, the fracture hits from P3E to the Codell fiber well P3Bf show fracture connection within Codell. The horizontal distance between the two Codell wells P3Bf and P3E is around 1,270 ft, suggesting the stimulated fracture half-length of the Codell wells is longer than the current spacing.



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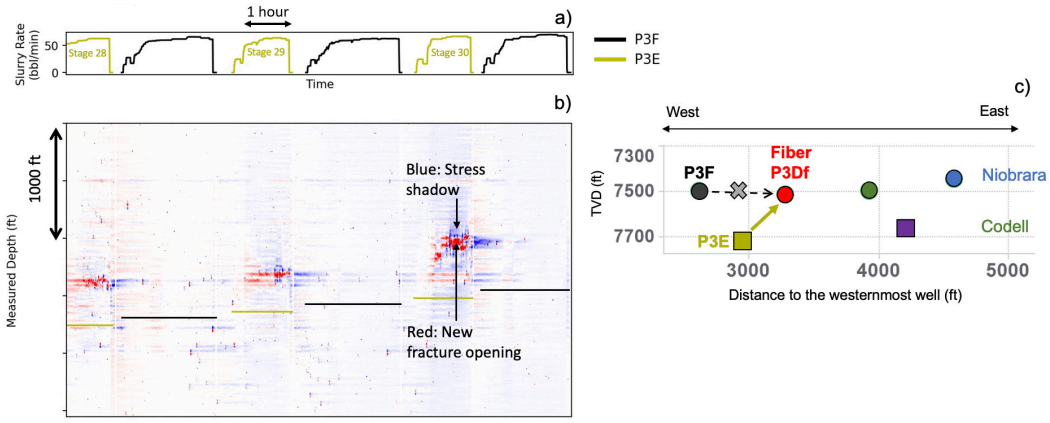


**Fig. 10—(a) The pumping curves of P3E (yellow curve) and P3F (black curve); (b) the recorded response of injectors' (P3E and P3F) zipper fracturing in the fiber well P3Df. Blue color indicates fiber compressing, whereas red refers to extending. The horizontal lines above the LF-DAS signal illustrate the relative perforation depth from the two injectors; (c) the diagrammatic cross-sectional relative position of wells on Pad 3. LF-DAS data show fewer connections within Niobrara but a much stronger connection between Codell and Niobrara.**

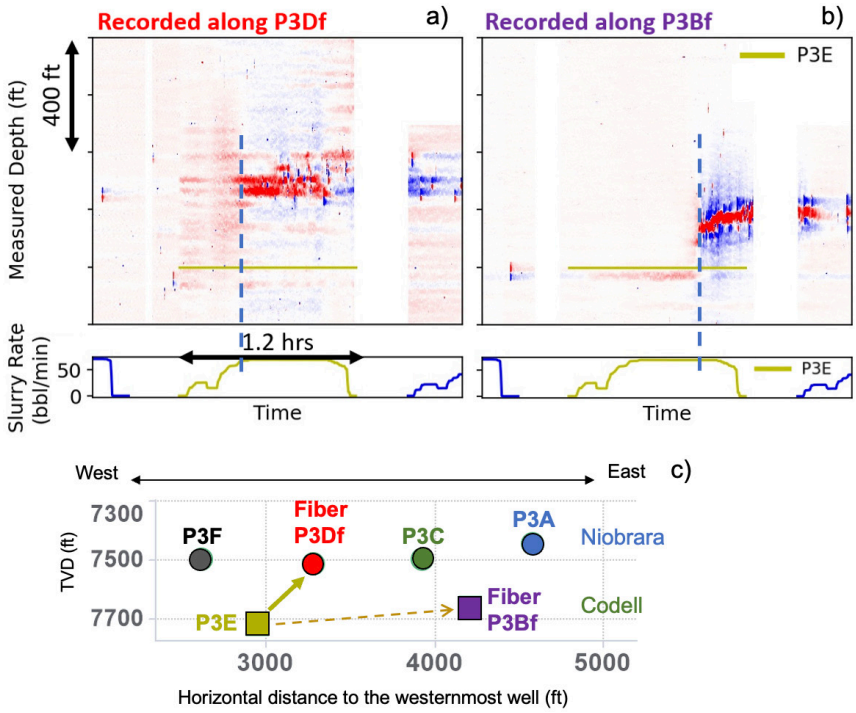


**Fig. 11—(a) The Niobrara well P3Df's LF-DAS data show fracture hit signals from the Codell well P3E (yellow), implying Niobrara-Codell communication; (b) the Codell well P3Bf's LF-DAS data identified fracture hits from the Codell well P3E, showing connection within Codell; (c) the connection summary is illustrated in the side view of Pad 3 wells.**

There are two differences between **Figs. 11a and 11b**. First, during the 1.2 hours of P3E's pumping period, when the first fracture hit was observed in both monitoring wells (P3Df and P3Bf), **Fig. 11b** illustrates a later fracture arrival time than **Fig. 11a**. The blue vertical dash lines on **Figs. 11a and 11b** record the timepoint when the first fracture hit was observed. To quantify the observation, the amount of slurry that has been injected so far was divided by the total amount of slurry that was injected into the entire stage. It was calculated that 16% of the slurry was injected when the first fracture hit was observed in **Fig. 11a**, while the first fracture hit in **Fig. 11b** corresponds to approximately 70% of the slurry. It is because the Codell monitor well P3Bf is farther apart from the injector P3E that fractures need to propagate longer and thus arrive later. Second, opposite from **Fig. 11a**, **Fig. 11b** shows that there is no signal polarity flip at the time of pumping stops, indicating fractures at the monitor well location P3Bf have already lost connection to the injector P3E during stimulation. The earlier fracture hit time and polarity flip in **Fig. 11a** correspond with strong connection between Niobrara and Codell formations during stimulation. Polarity flip indicates better fracture connectivity between the two wells, at least during injections. When compared with later fracture hit, earlier fracture hit means hydraulic fluid propagates for a longer period, hence it creates a hydraulic fracture much longer than the current well spacing. When assuming the hydraulic fracture filled with proppant is a constant percentage (e.g., less than 10%) of the entire hydraulic fracture filled with fluid, earlier fracture hit implies a longer proppant-filled hydraulic fracture. When a

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greater percentage of the well spacing is filled with proppant, the two wells that are connected by this hydraulic fracture are more likely to interfere with each other during the production stage.

## Comparison between Production and LF-DAS Analysis

**Table 4** summarizes the communication results of production and LF-DAS analysis. The direct correlation between the production interference results **(Fig. 9)** and the LF-DAS fracture hit signals implies that there is strong communication between Niobrara and Codell adjacent wells, during both stimulation period and production period. Additionally, both LF-DAS and production data show negligible or no communication between Niobrara wells. The LF-DAS data suggest that the half-length of stimulated hydraulic fractures in Niobrara wells is less than the current spacing of 650 ft. For Codell wells, though fracture hits within Codell are identified from LF-DAS, the phenomenon of no polarity flip in LF-DAS signals when pumping stops denotes weak connections during stimulation. This is related to the observation in **Fig. 9** that no production communication within Codell is observed. It suggests that the stimulated hydraulic fracture half-length of Codell wells is longer than the current spacing of 1,270 ft, which is much greater than that of Niobrara wells. Given that the half-length of drained hydraulic fractures in Codell wells is less than 635 ft, the ratio of drained hydraulic fracture half-length to the stimulated hydraulic fracture half-length is less than 50% for Codell wells.

| Communication Results | Production Data | LF-DAS Data |
|---|---|---|
| Niobrara-Codell (typical vertical spacing: 190 ft) | Up to 80% of production diversion means strong communication | Polarity flip and earlier fracture hit time suggest strong connection |
| Within Niobrara (typical horizontal spacing: 650 ft) | Negligible production interference indicates weak communication | No observed fracture hit indicates no connection |
| Within Codell (typical horizontal spacing: 1,270 ft) | No production communication is observed | No polarity flip and later fracture hit time imply weak connection during stimulation |

Table 4—Communication summary of production and LF-DAS results.

**Table 4** shows strong cross-formation connection but less inner-formation connection. This may be attributed to the difference in well spacing. The vertical spacing (190 ft) is significantly less than the horizontal spacing (650ft for Niobrara and 1,270ft for Codell). Additionally, **Table 4** shows that the half-length of hydraulic fractures is shorter in Niobrara than in Codell. This may be explained by the different natural fracture densities and in-situ stress conditions in the two formations, which affect the hydraulic fracture geometry. In this study area, Niobrara has a significantly higher natural-fracture density than Codell. The dense natural-fracture network may cause the hydraulic fractures in Niobrara to form more complex and shorter fracture systems. Furthermore, Niobrara formation may have different in-situ stress condition than the Codell formation in our study area, which is supported by the different fracture orientations in the two formations.

## Conclusions

In this study, we analyzed well interference in the unconventional DJ Basin reservoirs by integrating production data with LF-DAS data. The following conclusions are drawn:

- Polarity flip and earlier fracture hit times observed from LF-DAS signals between Niobrara and Codell formations during stimulation indicate strong interformation communication, which is consistent with strong well interference during production. Interwell fractures could divert up to 80% of a shut-in well's production to its offset well in the adjacent formation.
- There is negligible fracture connection in Niobrara wells during both stimulation and production. However, within Codell wells, fracture connection exists during stimulation but does not exist during production.
- The stimulated hydraulic fracture half-length of Codell wells is significantly longer than that of Niobrara wells. This confirms that the current well spacing design is qualitatively reasonable: Codell wells should be spaced more widely than Niobrara wells.
- The drained hydraulic fracture half-length of Codell wells is less than 635 ft, while the stimulated fracture half-length of Codell wells is greater than 1,270 ft. This indicates that less than 50% of the stimulated hydraulic fractures' half-length is drained by current Codell wells.

## Nomenclature

$b$ = decline exponent, dimensionless
$D_i$ = initial decline rate, days$^{-1}$
$q_i$ = initial production rate, B/D
$q_t$ = production rate at time $t$, B/D
$t$ = time, days

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